# Ritika Mehta

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## EDUCATION

**Carnegie Mellon University**, Pittsburgh PA *May 2025*
Bachelor of Science in Neuroscience, GPA: 3.67/4.0
Concentration in Computational Neuroscience, Minor in Computer Science (planned)

**Relevant Coursework:** Introduction to Machine Learning, Introduction to Computer Systems, Principles of Imperative Computation, Functional Programming, Probability Theory for Computer Scientists, Matrices and Linear Transformations

## SKILLS

**Programming Languages**: Python, C, HTML/CSS, LaTeX
**Frameworks**: NumPy, OpenCV, Pandas, LangChain

## EXPERIENCE

**Principles of Computing Teaching Assistant, Carnegie Mellon University** *August 2023 –*

- Teach introductory reasoning skills, algorithms, and Python concepts to 20+ students weekly

**Artificial Intelligence and Machine Learning Intern, StratiFi** *May 2023 – July 2023*

- Developed a financial advisor chatbot using the LangChain framework and GPT language model, enabling personalized financial guidance and recommendations for StratiFi's clients
- Created data analysis and code understanding agents with vector databases such as ChromaDB and Deeplake to analyze financial data and provide actionable insights
- Implemented intent analyzing and verification features using prompt engineering & memory to enable smoother user experience & more accurate responses from chatbot

## PROJECTS

**ENAiBLE Innovation Summer Research Program, Carnegie Mellon University** *May 2023 – August 2023*

- Conducted comprehensive background research, consumer interviews, and idea generation to develop solution focused on enhancing the in-store grocery shopping experience
- Utilized Figma to prototype "Grocer Greg," a personal AI assistant that provides live notifications and personalized recommendations to users
- Demonstrated strong teamwork, research, and design skills throughout the program and successfully pitched the final idea in a webinar to a 70+ audience of industry professionals

**AI Robo Therapist, Cognitive Robotics Project** *Spring 2023*

- Programmed a AI Cozmo Robo-therapist using GPT 3.5 and the Cozmo SDK
- Utilized prompt engineering techniques to implement sentiment analysis and human-like conversational features
- Implemented face tracking, facial emotional recognition and animation in response to user queries for optimal engagement

**Introduction to Computer Systems Labs** *Spring 2023*

- Malloc: Implemented dynamic memory allocator in C using segregated lists that supports malloc, free, calloc and realloc function calls
- Cache: Implemented a cache simulator and applied optimization strategies to minimize cache misses
- Attack: Exploited buffer overflow attacks using code injection attacks and return-oriented programming attacks
- Shell: Implemented unix shell with job control and fg, bg commands using control, signals, and signal handling techniques

**Fundamentals of Programming and Computer Science Term Project** *Spring 2022*

- Coded cooking game '*Dhabha Dash'* in Python, implemented path finding algorithm DFS to move player and customers with player, customer and object tracking
- Developed AI algorithms that calculate customers' experiences, tip players, cook food with automated chef and timer

## ACTIVITIES

**Girls Who Code, Activity Committee** *January 2023 –*

- Collaborate with other team members to develop an engaging and effective curriculum for Python classes for young girls